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SQL PROJECT

Nova Retail Group Database Analysis

Business Intelligence & Data Analytics

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Table of Contents

1. Project Overview

Company Background

Nova Retail Group is a growing retail company operating across multiple regions in South Africa. The company sells consumer products through both physical stores and an online platform. Management needs your help analyzing their data to make better business decisions.

Your Role

You are a Junior Data Analyst tasked with analyzing sales data, customer behavior, and product performance. You will write SQL queries to answer specific business questions and provide insights to management.

Learning Objectives

By completing this project, you will demonstrate proficiency in:

- Basic SQL queries (SELECT, WHERE, ORDER BY)
- Aggregate functions (COUNT, SUM, AVG, MIN, MAX)
- JOINs (INNER JOIN, LEFT JOIN, RIGHT JOIN)
- GROUP BY and HAVING clauses
- Subqueries and CTEs (Common Table Expressions)
- Window functions (RANK, ROW_NUMBER, PARTITION BY)
- Date functions and time-based analysis
- Business intelligence and data interpretation

2. Database Schema

Database Name: NovaRetailDB

Table 1: Products

Contains product catalog information (30 products).

Column	Data Type	Description
ProductID (PK)	VARCHAR(10)	Unique product identifier
ProductName	VARCHAR(100)	Product name
Category	VARCHAR(50)	Product category
UnitPrice	DECIMAL(10,2)	Selling price

CostPrice	DECIMAL(10,2)	Cost to acquire product

Table 2: Customers

Contains customer information (500 customers).

Column	Data Type	Description
CustomerID (PK)	VARCHAR(10)	Unique customer ID
FirstName, LastName	VARCHAR(50)	Customer name
Region	VARCHAR(20)	North, South, East, West
Channel	VARCHAR(20)	Online or Store
JoinDate	DATE	Date joined

Table 3: Sales

Contains transaction data (2,500 records). This is the main fact table.

Column	Data Type	Description
OrderID (PK)	VARCHAR(20)	Unique order ID
OrderDate	DATE	Transaction date
CustomerID (FK)	VARCHAR(10)	Links to Customers
ProductID (FK)	VARCHAR(10)	Links to Products
Quantity, UnitPrice	INT, DECIMAL	Order details
TotalSales, Profit	DECIMAL(10,2)	Financial metrics
Channel	VARCHAR(20)	Online or Store

Table 4: CustomerFeedback

Contains customer satisfaction surveys (1,000 records).

Column	Data Type	Description

FeedbackID (PK)	VARCHAR(20)	Unique feedback ID
OrderID, CustomerID	VARCHAR	Foreign keys
Rating	INT	1-5 star rating
Satisfaction	VARCHAR(50)	Satisfaction level
RecommendLikelihood	INT	1-10 NPS score

3. Setup Instructions Create an ERD diagram here

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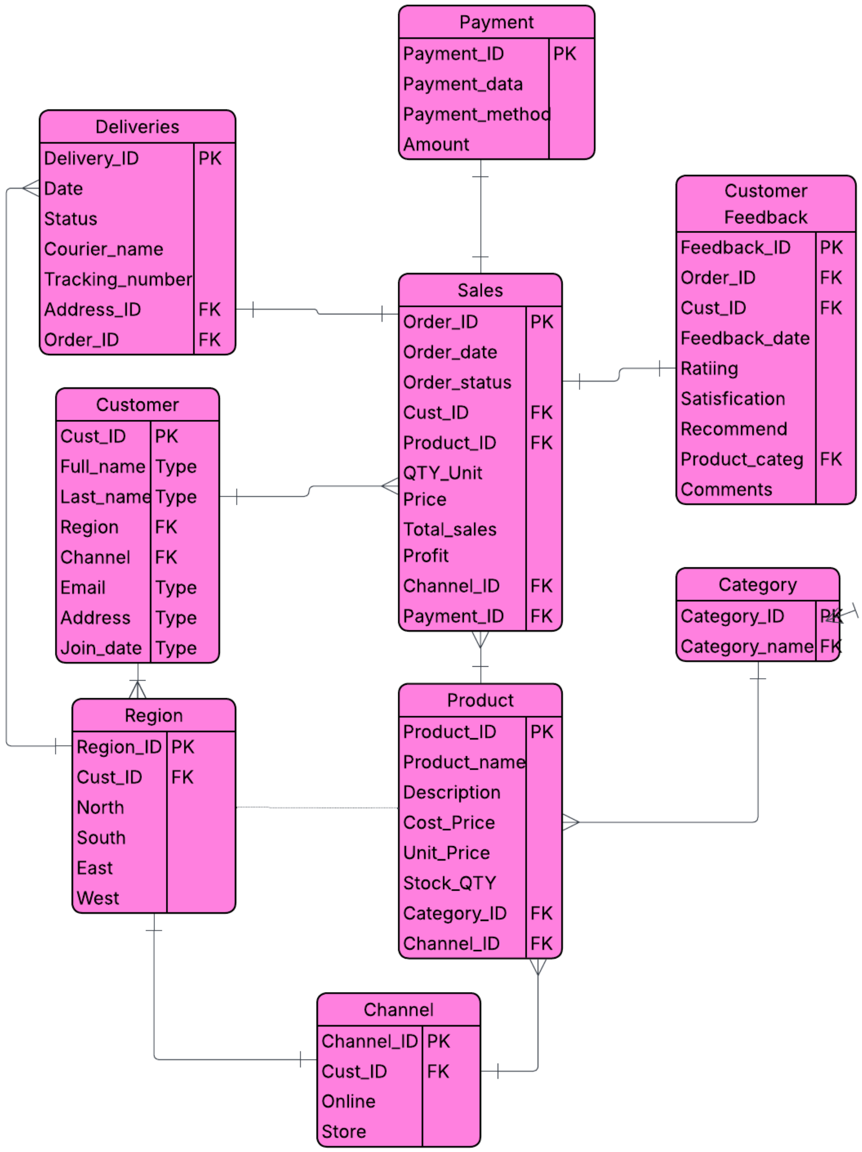
Nova Retail ERD



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Step 1: Create Database(Ignore)

Run the SQL_Complete_Database_Setup.sql script to create all tables.

Step 2: Import Data

Import the CSV files for Customers, Sales, and CustomerFeedback tables using your database tool.

Step 3: Verify Import

Run the following query to verify all data is loaded:

```
SELECT 'Products' AS TableName, COUNT(*) AS Records FROM Products UNION ALL
SELECT 'Customers', COUNT(*) FROM Customers UNION ALL SELECT 'Sales',
COUNT(*) FROM Sales UNION ALL SELECT 'CustomerFeedback', COUNT(*) FROM
CustomerFeedback;
```

Expected Results:

- Products: 30
- Customers: 500
- Sales: 2,500
- CustomerFeedback: 1,000

4. Part 1: Basic SQL Queries (Beginner)

Write SQL queries to answer the following questions. Test each query and paste the results.

Question 1.1: Product Catalog (5 points)

List all products in the Electronics category. Show ProductID, ProductName, and UnitPrice. Order by price from highest to lowest.

Your SQL Query:

```
select productID, productName, UnitPrice
from products
order by unitprice desc
```

Question 1.2: Customer Count (5 points)

How many customers are in each region? Show Region and the count of customers. ***Your SQL Query:***

```
select LastName, region,
count(customerID) as total
from customers
where region = 'North'
or region = 'South'
or region = 'East'
or region = 'West'
group by LastName, region
```

Question 1.3: Recent Orders (5 points)

Show the 10 most recent orders. Display OrderID, OrderDate, and TotalSales. ***Your SQL Query:***

```
select orderID, orderDate, TotalSales
from sales
order by orderDate desc
limit 10
```

Question 1.4: Affordable Products (5 points)

Find all products with a UnitPrice less than R1000. Show ProductName, Category, and UnitPrice.

Your SQL Query:

```
select ProductName, Category, UnitPrice
from products
where UnitPrice < 1000
```

Question 1.5: Customer Satisfaction Summary (5 points)

Count how many feedback responses exist for each Satisfaction level. Order by count descending.

Your SQL Query:

```
select Satisfaction, count(*) as total
from customer_feedback
group by Satisfaction
```

5. Part 2: Intermediate SQL Queries

These questions require JOINs, aggregate functions, and GROUP BY clauses.

Question 2.1: Sales by Category (10 points)

Calculate total sales revenue for each product category. Show Category, Total Revenue, and Total Profit. Order by revenue descending.

Your SQL Query:

```
select category, sum(totalsales) as total_revenue,
sum(profit) as total_profit
from products
inner join sales on products.productid = sales.productid
group by category
```

```
order by total_revenue desc
```

Question 2.2: Top Customers (10 points)

Find the top 5 customers by total purchase amount. Show CustomerID, Customer Name (FirstName + LastName), and Total Spent. Hint: Use CONCAT or + to combine names.

Your SQL Query:

```
with
TotalSales as (
select sales.customerid, CONCAT('firstname','', 'lastname') as
customer_name, sum(totalsales) as totalSales
from sales
group by customerID
),
highest_purchase as (
select sales.customerid, CONCAT('firstname', 'lastname') as
customer_name, totalsales
from sales
inner join customers on sales.customerid = customers.customerid
where totalsales = (select max(totalsales) from totalsales)
)
select customerid,CONCAT('firstname', '', 'lastname') as
customer_name, totalsales
from totalsales
union all
select customerid,CONCAT('firstname', '' ,'lastname') as
customer_name, totalsales
from highest_purchase
order by totalsales desc
limit 5
```

Question 2.3: Monthly Sales Trend (10 points)

Show total sales by month for 2024. Display Year, Month (as number), and Total Sales. Order chronologically.

Your SQL Query:

```
select YEAR(OrderDate) as year, MONTH(OrderDate)
as month, sum(TotalSales) as total_sales
```

```sql
from sales
where YEAR(OrderDate) = 2024
group by year, month
order by year, month
```

Question 2.4: Channel Performance (10 points)

Compare Online vs Store performance. Show Channel, Number of Orders, Average Order Value, and Total Revenue.

Your SQL Query:
```sql
select Channel, count(OrderID) as total_orders,
AVG(TotalSales) as avg_order_value, sum(TotalSales) as total_revenue
from sales
group by Channel
order by total_revenue desc
```

Question 2.5: Product Performance with Ratings (10 points)

For each product category, show the average customer rating. Include Category, Average Rating, and Number of Reviews. Only show categories with at least 50 reviews.

Your SQL Query:
```sql
select Category, AVG(Rating) as avg_rating,
count(*) as total_reviews
from customer_feedback
inner join products on customer_feedback.ProductCategory =
products.Category
group by Category
having total_reviews >= 50
order by avg_rating desc
```

6. Part 3: Advanced SQL Queries

These questions require subqueries, CTEs, window functions, or complex JOINs.

Question 3.1: Best Selling Product per Category (15 points)

Find the product with the highest total sales in each category. Show Category, ProductName, and Total Revenue for that product. Use a window function or subquery. ***Your SQL Query:***

```
select category, ProductName, sum(TotalSales) as total_revenue
from products
inner join sales on products.productid = sales.productid
group by category, ProductName
order by total_revenue desc
```

Question 3.2: Customer Lifetime Value (15 points)

Create a query showing each customer's total purchases, number of orders, and average order value. Also include their region and primary channel. Only show customers with more than 3 orders. Order by total purchases descending. ***Your SQL Query:***

```
select customers.CustomerID, customers.Region, customers.Channel,
count(sales.OrderID) as total_orders, AVG(sales.TotalSales) as
avg_order_value, sum(sales.TotalSales) as total_purchase
from customers
inner join sales on customers.CustomerID = sales.CustomerID
group by customers.CustomerID, customers.Region, customers.Channel
having total_purchase >= 3
order by total_purchase desc
```

Question 3.3: Profit Margin Analysis (15 points)

Calculate the profit margin percentage for each product. Show ProductName, Category, Total Sales, Total Profit, and Profit Margin %. Order by profit margin descending. Formula: (Profit / Sales) * 100 ***Your SQL Query:***

```
select
    sales.ProductName,
    products.category,
    SUM(sales.TotalSales) as totalsales,
    SUM(sales.Profit) as totalprofit,
    (SUM(sales.Profit)/SUM(sales.TotalSales))*100 as profit_margin
from sales
inner join products on sales.productid = products.productid
group by sales.ProductName, products.category
order by profit_margin desc
```

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Question 3.4: Year-over-Year Growth (15 points)

Compare sales between 2023 and 2024. Show the total sales for each year and calculate the growth percentage. Use a CTE or subquery.

Your SQL Query:

```
with sales_2023 as (
    select sum(TotalSales) as total_sales_2023
    from sales
    where YEAR(OrderDate) = 2023
),
sales_2024 as (
    select sum(TotalSales) as total_sales_2024
    from sales
    where YEAR(OrderDate) = 2024
)
select
    s23.total_sales_2023,
    s24.total_sales_2024,
    ((s24.total_sales_2024 - s23.total_sales_2023) /
s23.total_sales_2023) * 100 as growth_percentage
from sales_2023 s23
cross join sales_2024 s24
```

Question 3.5: Regional Performance Ranking (15 points)

Rank regions by total sales. Use the RANK() or ROW_NUMBER() window function. Show Region, Total Sales, Total Orders, and Rank.

Your SQL Query:

```
with regional_sales as (
    select Region, count(OrderID) as totalorders, sum(TotalSales) as
totalsales
    from sales
    inner join customers on sales.CustomerID = customers.CustomerID
    group by Region
)
select
    Region,
    totalsales as total_sales,
```

```
            totalorders as total_orders,
    Rank() over (
        partition by Region
        order by totalsales desc
    ) as rank
from regional_sales
```

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7. Part 4: Business Intelligence Questions

Answer these strategic questions using SQL queries and provide business insights.

Question 4.1: Customer Satisfaction vs. Repeat Purchases (20 points)

Analyze whether highly satisfied customers (rating 4-5) make more repeat purchases than less satisfied customers (rating 1-3). Show satisfaction level groups, average number of orders per customer, and total customers in each group.

Your SQL Query:

```
with customer_orders as (
    select
        cf.CustomerID,
        cf.Rating,
        count(s.OrderID) as total_orders
    from customer_feedback cf
    left join sales s on cf.CustomerID = s.CustomerID
    group by cf.CustomerID, cf.Rating
)
select
    case
        when Rating between 4 and 5 then 'Highly Satisfied'
        when Rating between 1 and 3 then 'Less Satisfied'
        else 'Other'
    end as satisfaction_level,
    avg(total_orders) as avg_orders_per_customer,
    count(distinct CustomerID) as total_customers
from customer_orders
group by satisfaction_level
```

Business Insight (2-3 sentences): From the above query results, it is visible that highly satisfied customers have high average orders per customer, which is 8.7. Whereas less satisfied customers have an average of 6.8 orders per customer.

Question 4.2: Discount Effectiveness (20 points)

Examine the relationship between discount percentage and profit. Create discount bands (0%, 1-10%, 11-20%, 21-30%) and show total sales, total profit, and profit margin for each band.

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Your SQL Query:
```
with discount_bands as (
    select
        case
            when DiscountPercent = 0 then '0%'
            when DiscountPercent between 0.01 and 0.10 then '1-10%'
            when DiscountPercent between 0.11 and 0.20 then '11-20%'
            when DiscountPercent between 0.21 and 0.30 then '21-30%'
            else 'Other'
        end as discount_band,
        sum(TotalSales) as totalsales,
        sum(Profit) as totalprofit,
        (sum(Profit)/sum(TotalSales))*100 as profit_margin
    from sales
    group by discount_band
)
select
    discount_band,
    totalsales,
    totalprofit,
    profit_margin
from discount_bands
order by discount_band
```

Business Insight (2-3 sentences): In 0% discount, there's the highest profit margin because they are selling at full price. At 10% discount, they can boost sales volume while keeping profit margin relatively healthy, hence there are more satisfied customers than dissatisfied. Then, at 20% profit can be weak, not good for business.

Question 4.3: Product Portfolio Optimization (20 points)

Identify underperforming products (bottom 5 by total revenue) and analyze whether they should be discontinued. Consider sales volume, profit margin, and customer ratings. ***Your SQL Query:***
```
with
product_sales as (
    select
        p.ProductID,
```

```
        p.ProductName,
        p.Category,
        p.UnitPrice,
        p.CostPrice,
        sum(TotalSales) as totalsales,
        sum(Profit) as totalprofit,
        avg(cf.Rating) as avg_rating
    from products p
    left join sales s on p.ProductID = s.ProductID
    left join customer_feedback cf on s.CustomerID = cf.CustomerID
    group by p.ProductID, p.ProductName, p.Category, p.UnitPrice,
p.CostPrice
)
select
    ProductID,
    ProductName,
    Category,
    UnitPrice,
    CostPrice,
    totalsales,
    totalprofit,
    avg_rating
from product_sales
order by totalsales asc
limit 5
```

Business Recommendation (3-4 sentences): *From* the results shown, sales are improving from 84k to 113k, showing consistent demand. Profits are also increasing, whereas the profit margins appear to be stable. Which means growth is happening, and there's also room for improvement in areas such as product quality and customer experience. So, inclosing the underperforming product should not be discontinued.

8. Part 5: Executive Management Report

Create a comprehensive report for Nova Retail Group's executive team. Your report should include SQL queries and written analysis.

Report Requirements (50 points total)

Section 1: Overall Business Performance (15 points)

Provide a summary of key business metrics:

- Total revenue and profit for 2023-2024
- Overall profit margin
- Total number of orders and customers
- Average order value

Your SQL Query:

```
with sales_2023_2024 as (
    select
        sum(TotalSales) as total_revenue,
        sum(Profit) as total_profit,
        count(DISTINCT OrderID) as total_orders,
        count(DISTINCT CustomerID) as total_customers,
        avg(TotalSales) as avg_order_value
    from sales
    where OrderDate between '2023-01-01' and '2024-12-31'
)
select
    total_revenue,
    total_profit,
    (total_profit/total_revenue)*100 as profit_margin,
    total_orders,
    total_customers,
    avg_order_value
from sales_2023_2024
```

Section 2: Top 3 Insights (20 points)

Identify and explain the three most important insights from your analysis. Each insight must be supported by SQL query results.

Insight #1: From these business results, the overall conclusion is very positive. The products are financially strong and valuable. With high revenue and profitability, the business is generating substantial returns. Strong profit margins, showing efficient cost management, and strong pricing power. Customers are showing their loyalty by placing more orders; this repeat buying shows customer satisfaction. This business is in a strong position.

Supporting SQL Query:

```
with sales_2023_2024 as (
    select
        sum(TotalSales) as total_revenue,
        sum(Profit) as total_profit,
        count(DISTINCT OrderID) as total_orders,
        count(DISTINCT CustomerID) as total_customers,
        avg(TotalSales) as avg_order_value
    from sales
    where OrderDate between '2023-01-01' and '2024-12-31'
)
select
    total_revenue,
    total_profit,
    (total_profit/total_revenue)*100 as profit_margin,
    total_orders,
    total_customers,
    avg_order_value
from sales_2023_2024
```

Insight #2: The Nova Retail Store shows a 10% increase growth when we compare 2023-2024, demonstrating resilience and consistent performance. Moreover, the focus should be on accelerating growth by expanding the customer base, improving ratings, and leveraging marketing to push beyond 10%. The next step is to identify what drove the growth in 2023-2024 period. From my analysis, repeat buyers, especially online, played a big role in increasing sales, especially in the electric category.

Supporting SQL Query:

```sql
with sales_2023 as (
    select sum(TotalSales) as total_sales_2023
    from sales
    where YEAR(OrderDate) = 2023
),
sales_2024 as (
    select sum(TotalSales) as total_sales_2024
    from sales
    where YEAR(OrderDate) = 2024
)
select
    s23.total_sales_2023,
    s24.total_sales_2024,
    ((s24.total_sales_2024 - s23.total_sales_2023) / 
s23.total_sales_2023) * 100 as growth_percentage
from sales_2023 s23
cross join sales_2024 s24
```

Insight #3: From the below query are the results for the products with the highest sales in each category. It is visible that electronics and home appliances make the highest number of sales, meaning the customers buy electronics and home appliances more than other things. So, my suggestion would be that the store should stock more in those categories both online and in physical stores.

Supporting SQL Query:

```sql
select category, ProductName, 
sum(TotalSales) as total_revenue
from products
inner join sales on products.productid = sales.productid
group by category, ProductName
order by total_revenue desc
```

Section 3: Strategic Recommendations (15 points)

Based on your analysis, provide 3-5 actionable recommendations for management. Each recommendation should be data-driven.

Actionable recommendations.

1. Management should focus resources on high performing electronics, optimize the appliance portfolio, and critically assess lifestyle goods, focus on refinement by scaling what works, and trimming what doesn't. Expanding the customer base to unlock higher growth.
2. Management should prioritize online expansion while transforming stores into valuable adding hubs. Online is the revenue engine, but physical stores can strengthen brand trust, customer engagement, and upselling opportunities.
3. In closing management should also focus more on marketing, especially on social media platforms such as Facebook, Instagram, TikTok, and WhatsApp to attract more customers.

9. Submission Guidelines

What to Submit

- This completed document with all SQL queries and answers
- A separate .sql file containing all your queries organized by section
- Screenshots of query results for Part 5 (Management Report)

Grading Criteria

Section	Points	Criteria
Part 1: Basic SQL	25	Correctness, syntax
Part 2: Intermediate SQL	50	JOINs, aggregations
Part 3: Advanced SQL	75	Subqueries, CTEs, windows
Part 4: Business Intelligence	60	Analysis, insights
Part 5: Management Report	50	Strategic thinking
TOTAL	**260**	

Tips for Success

- Test each query multiple times before submitting
- Use meaningful column aliases to make results clear
- Format your SQL code properly (indentation, line breaks)
- Add comments to complex queries explaining your logic
- Round decimal numbers appropriately (2 decimal places for currency) •   Think like a business analyst when interpreting results

Good Luck!